

CDS 101 – Final Project Report

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1. Problem Definition

This project will analyze the relationship between tuition costs and graduation rates based on a dataset from the US Department of Education. Specifically, whether institutions with higher net prices tend to have higher graduation rates. Graduation rate is used as a proxy for student success, while net price represents the average cost students pay after financial aid. This will be beneficial to incoming college students who want to see if the higher cost is worth it. College graduates may also be interested in seeing if their college tuition was worth it, or not, in addition to their experience after graduation.

We expect that colleges with higher net prices will have higher graduation rates. This expectation is based on the idea that more expensive institutions often have more financial resources, provide better academic support services, admit more academically prepared students, and have lower student-to-faculty ratios.

However, the relationship may not be extremely strong. Net price alone does not capture all factors influencing graduation rates. Other important variables include admission selectivity, student socioeconomic background, institutional type (public vs private), and Academic preparedness of students.

There is also a consideration that a more expensive college may have better education, so the graduation rates may be lower than a college one might consider “easier”; this relies on how well the education at a college is, which this model wouldn’t show.

2. Data Acquisition & Description

- **Name and source of the dataset(s).**

College.rds dataset from the instructor.

- **How you obtained the data (download link, API, etc.).**

Canvas download from CDS 101

- **Brief description of the variables / features (with units if relevant).**

There are many variables about college information, but we will only focus on:

C150_4 # 4-year graduation rate

NPT4_PUB # public schools

NPT4_PRIV # private schools

ADM_RATE # selectivity

CONTROL # public vs private

LOCALE # urban/rural

SATVRMID, SATMTMID # academic preparedness

- **Approximate size (rows, columns).**

Rows: 7,058 Columns: 1,977

- **Any ethical, privacy, or bias considerations (if applicable).**

There may be cultural and societal factors that are not accounted for, especially because there are divisions among race and sex. We are also going into the project with the assumption higher tuition may cause lower graduation rate, given a better education is more expensive and difficult.

3. Data Cleaning & Preprocessing

Describe the steps you took to make the data usable, for example:

- **Handling missing values (drop, impute, etc.).**

We dropped the rows that weren’t a graduation rate of 4 years/ 4 year institutions

- **Fixing types (dates, numeric vs. categorical).**

We dropped rows that did not contain any information about the graduation rates and net prices (C150_4, NPT4_PUB, NPT4_PRIV).

- **Removing outliers (if done).**

We didn’t remove any outliers because they are needed for analysis.

- **Creating new features (feature engineering).**

A new feature, net_price, was created by combining private and public institution net price.

- **Any scaling or encoding.**

No scaling was needed.

```
college_clean <- college %>%
  dplyr::filter(PREDDEG == 3) %>%
  dplyr::mutate(
    grad_rate = C150_4,
    net_price = dplyr::coalesce(NPT4_PUB, NPT4_PRIV),
    type = ifelse(CONTROL == 1, "Public", "Private")
  ) %>%
  dplyr::filter(
    !is.na(grad_rate),
    !is.na(net_price),
    net_price > 0
  ) %>%
  dplyr::select(INSTNM, grad_rate, net_price, type)
```

4. Exploratory Data Analysis (EDA)

- Summary statistics

```
summary(college_clean %>% select(grad_rate, net_price))
```

```
##      grad_rate      net_price
## Min.   :0.0000   Min.    : 923
## 1st Qu.:0.3761   1st Qu.:14748
## Median :0.5210   Median :19474
## Mean   :0.5155   Mean    :19977
## 3rd Qu.:0.6629   3rd Qu.:24339
## Max.   :1.0000   Max.    :51190
```

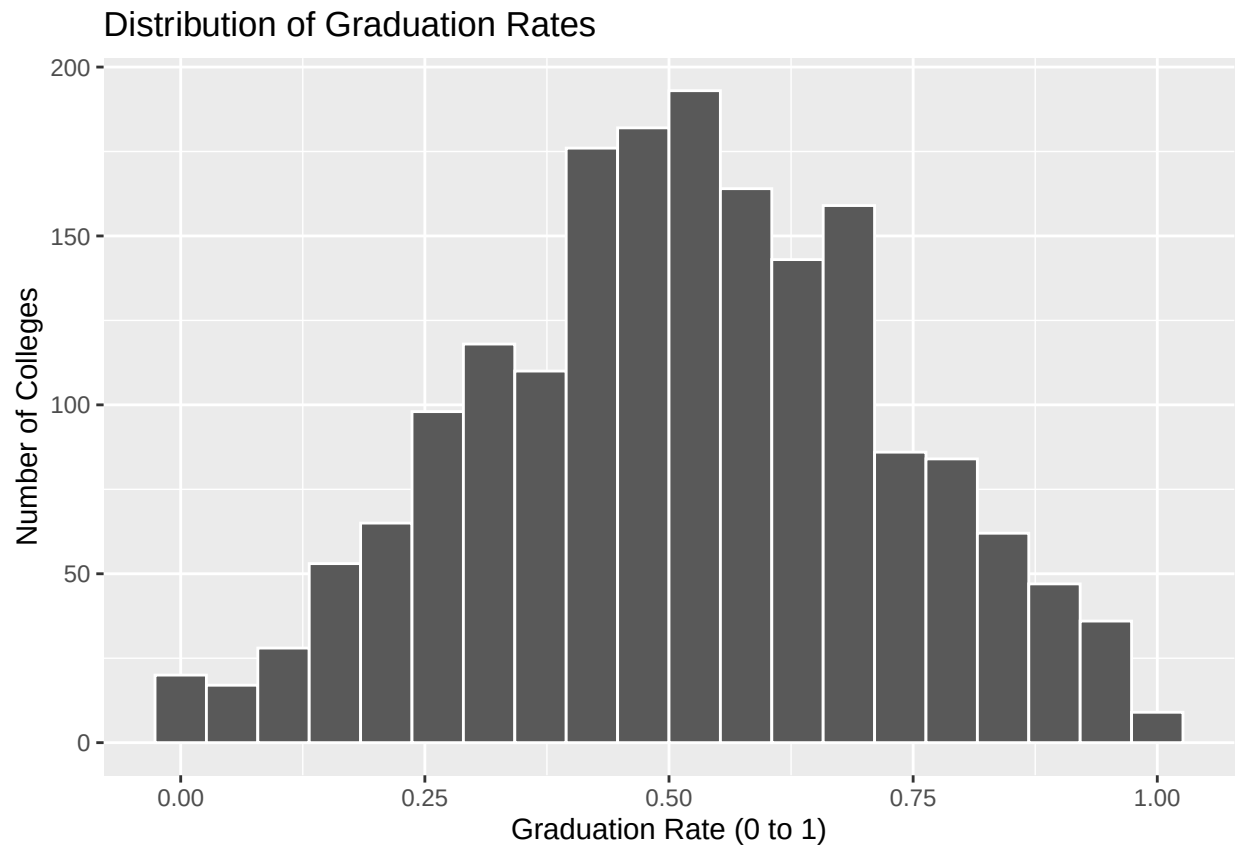
```
college_clean %>%
  summarise(
    mean_grad = mean(grad_rate),
    sd_grad = sd(grad_rate),
    mean_price = mean(net_price),
    sd_price = sd(net_price)
  )
```

```
## # A tibble: 1 x 4
##   mean_grad sd_grad mean_price sd_price
##   <dbl>    <dbl>    <dbl>    <dbl>
## 1     0.515    0.209    19977.    7712.
```

- The average graduation rate for the universities in our study is approximately 51.5%, with an average net price of about 19,977. These figures provide a clear baseline, showing that a typical student in our sample pays around 20,000 and has a little over a 50% chance of graduating on time. The standard deviations also indicate a wide variety of costs and outcomes across the 1,850 institutions we are analyzing.

- Visualizations (Histogram)

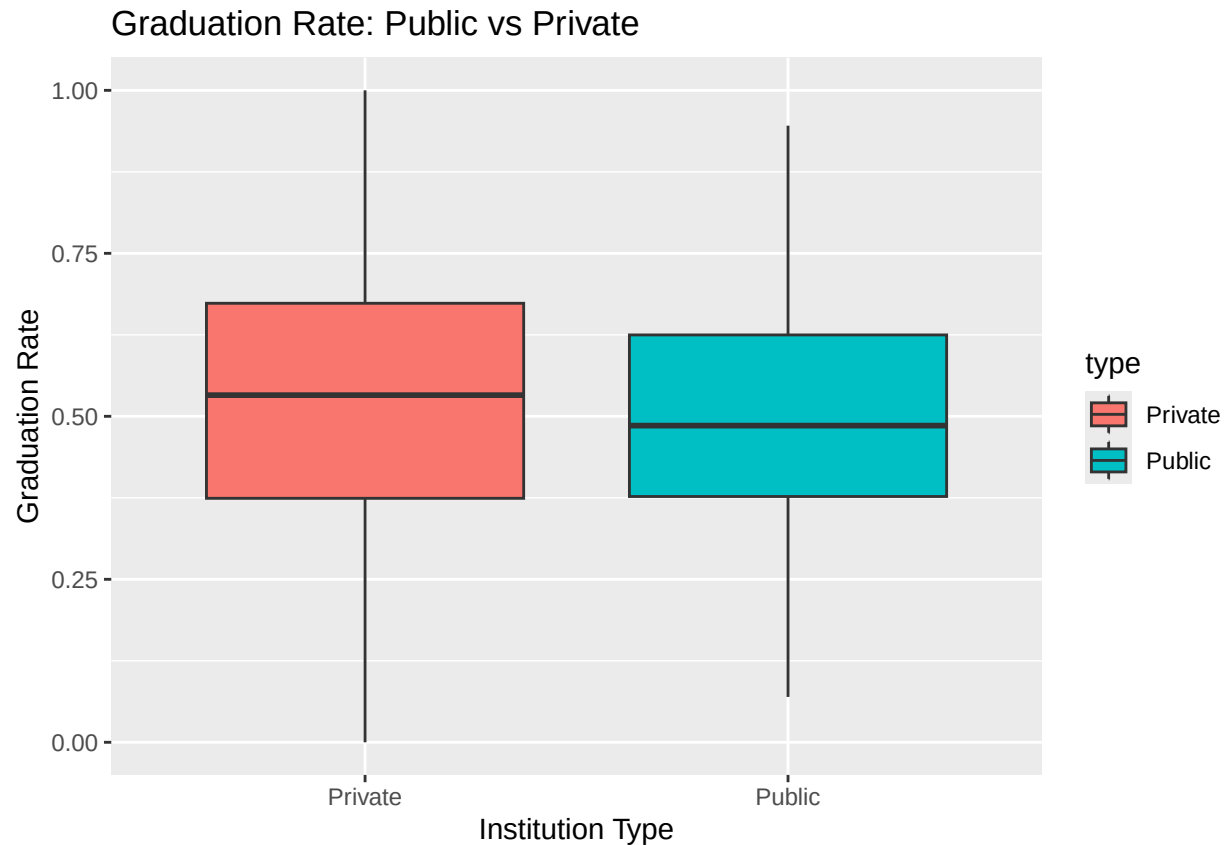
```
ggplot(college_clean, aes(x = grad_rate)) +
  geom_histogram(bins = 20, color = "white") +
  labs(
    title = "Distribution of Graduation Rates",
    x = "Graduation Rate (0 to 1)",
    y = "Number of Colleges"
  )
```



* The graduation rates follow a roughly symmetric, bell-shaped distribution centered around the 50% mark. This tells us that most colleges in our dataset have graduation rates in the middle range, while extremely high or low graduation rates are relatively rare. This normal distribution is helpful for our later modeling steps because it suggests the data is well-behaved.

- Visualizations (Boxplot)

```
ggplot(college_clean, aes(x = type, y = grad_rate, fill = type)) +
  geom_boxplot() +
  labs(title = "Graduation Rate: Public vs Private", x = "Institution Type", y = "Graduation Rate")
```

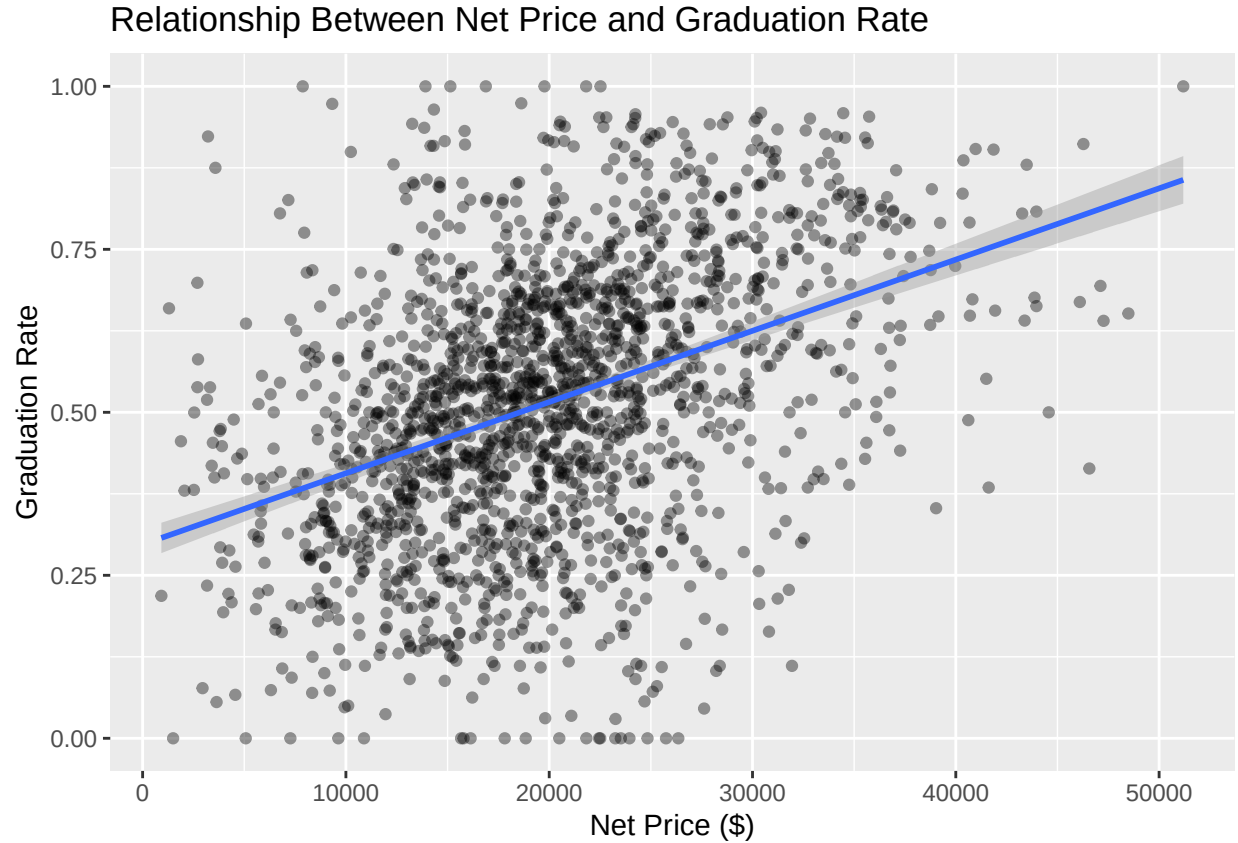


* When comparing institution types, private schools generally show a higher median graduation rate than public schools. The boxplot reveals that the middle 50% of private colleges typically perform better in terms of student success.

- Visualizations (Scatter Plot)

```
ggplot(college_clean, aes(x = net_price, y = grad_rate)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm") +
  labs(
    title = "Relationship Between Net Price and Graduation Rate",
    x = "Net Price ($)",
    y = "Graduation Rate"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



* There is a clear positive correlation between net price and graduation rate. It is suggesting that as the cost of tuition increases, the graduation rate tends to rise as well. This visual evidence directly supports our initial hypothesis and justifies using a linear regression model for the next stage of our project.

5. Visualization Quality and Storytelling

We used a histogram for graduation rates because it is the most effective way to show the frequency and distribution of a single numeric variable. A boxplot was chosen to compare graduation rates between public and private institutions as it clearly displays medians and the spread of data across categories. A scatter plot was used for the relationship between net price and graduation rate, because it is visualizing correlation between two numeric variables.

To improve clarity, we included descriptive titles and clear axis labels so that anyone can understand the units being measured. In the scatter plot, we used alpha transparency to address the issue of over-plotting which allowing the reader to see the density of the whole data points more clearly. We also added a linear regression line to the scatter plot to provide a clear visual guide for the trend.

6. Modeling Approach

Our project is framed as a regression task. Since our target variable (graduation rate) and our primary predictor (net price) are both continuous numerical values, we are looking to predict a specific numeric outcome rather than a category. For our baseline, we use the mean graduation rate of the dataset. This model assumes that every college has the same graduation rate (the average) regardless of its cost. We

used this baseline to evaluate whether our chosen regression model actually provides a more accurate and meaningful prediction.

We chose Simple Linear Regression as our primary model. Our Exploratory Data Analysis (EDA) revealed a roughly linear trend between net price and graduation rate. Linear regression provides a clear coefficient that tells us exactly how much the graduation rate is expected to change for every dollar increase in net price. Given that we are focusing on two key variables, a simple linear model prevents overfitting and provides a clear answer to our research question.

7. Model Implementation & Evaluation

```
library(modelr)
library(broom)
set.seed(123)
college_split <- resample_partition(college_clean, c(train = 0.8, test = 0.2))
train_data <- as_tibble(college_split$train)
test_data <- as_tibble(college_split$test)
```

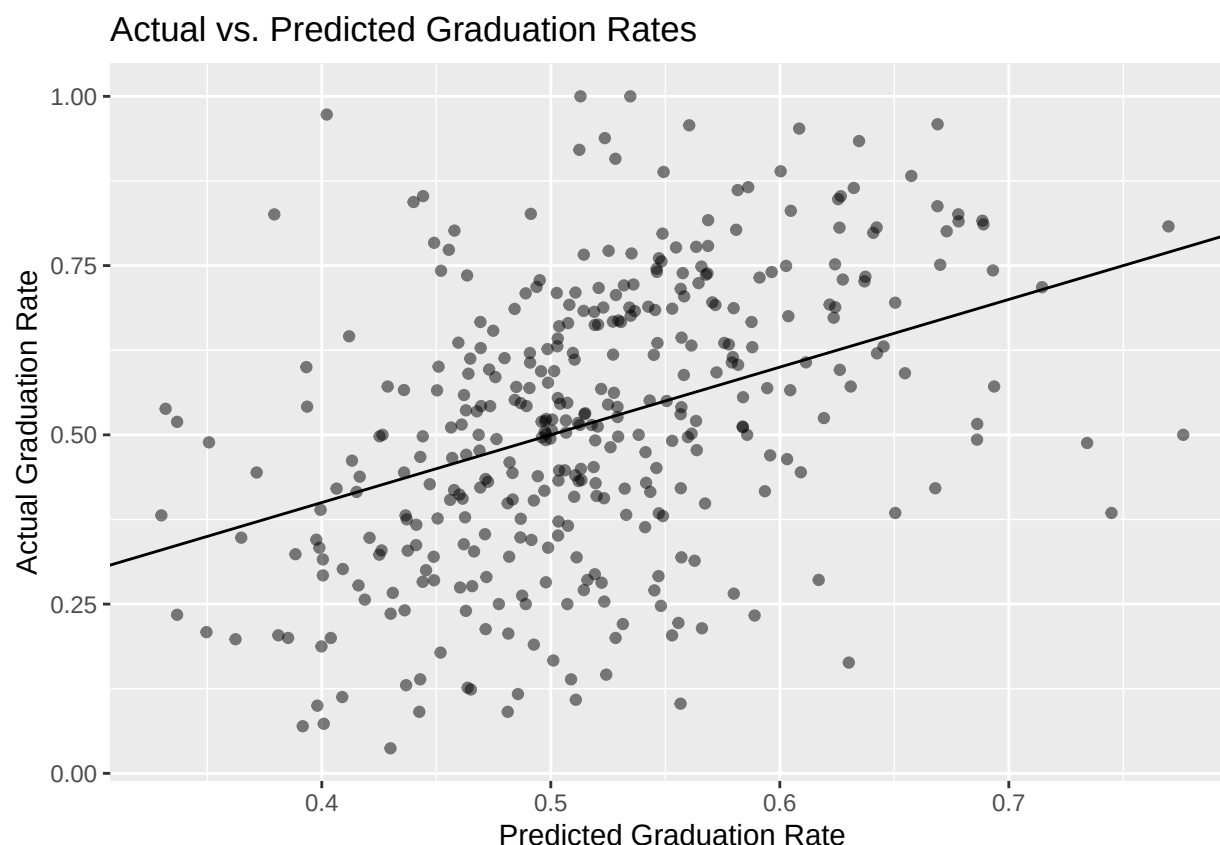
```
final_model <- lm(grad_rate ~ net_price, data = train_data)
```

```
test_data %>%
  summarise(
    RMSE = rmse(final_model, test_data),
    R_Squared = rsquare(final_model, test_data)
  )
```

```
## # A tibble: 1 x 2
##   RMSE R_Squared
##   <dbl>   <dbl>
## 1 0.185     0.187
```

```
test_results <- test_data %>%
  add_predictions(final_model)
```

```
ggplot(test_results, aes(x = pred, y = grad_rate)) +
  geom_point(alpha = 0.5) +
  geom_abline(slope = 1, intercept = 0) +
  labs(title = "Actual vs. Predicted Graduation Rates",
       x = "Predicted Graduation Rate",
       y = "Actual Graduation Rate")
```



For our model, we used `net_price` as the primary feature to predict the `grad_rate`. To ensure our model generalizes well to new data, we split our dataset into a training set (80%) to build the model and a testing set (20%) to evaluate its performance. We used a random seed to make sure our results are reproducible.

Our RMSE value shows the average difference between our model's prediction and the actual graduation rate. The R-squared value measures how much of the variation in graduation rates can be explained by the net price.

The 'Actual vs. Predicted' plot shows how closely our predictions align with the real-world data. While there is a positive trend, the spread of the data points suggests that while net price is a significant factor, it is not the only thing that determines student success. The line on the plot represents a perfect prediction, and seeing our points cluster around it confirms that our linear approach was appropriate for this analysis.

8. Conclusions & Recommendations

Our analysis confirms that there is a statistically significant positive relationship between a university's net price and its graduation rate. The R-squared value of 0.187 indicates that approximately 18.7% of the variation in graduation rates can be explained by the net price. While this shows that cost is a meaningful predictor, it also suggests that more than 80% of student success is determined by other factors not included in our simple model, such as student demographics, campus resources, or academic programs.

Our model's RMSE of 0.185 means that, on average, our predictions for graduation rates are off by about 18.5 percentage points. This level of error is expected when using a single predictor for a complex social outcome.

For prospective students and parents, our findings suggest that "you get what you pay for" to some extent, as more expensive institutions tend to have higher graduation rates. However, because price only explains a

small portion of the outcome, we recommend that students also evaluate other qualitative factors like faculty engagement and career services, rather than assuming a high price tag automatically guarantees success.

9. Code Quality & Reproducibility

The following R packages must be installed and loaded to execute the code successfully: tidyverse, modelr, and broom.

```
## R version 4.5.2 (2025-10-31)
## Platform: x86_64-apple-darwin20
## Running under: macOS Sequoia 15.7.5
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/lib/libRlapack.dylib; LAPACK
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] broom_1.0.12    modelr_0.1.11  lubridate_1.9.5 forcats_1.0.1
## [5] stringr_1.6.0   dplyr_1.1.4    purrr_1.2.1    readr_2.1.6
## [9] tidyr_1.3.2     tibble_3.3.1   ggplot2_4.0.1  tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] Matrix_1.7-4      gtable_0.3.6      compiler_4.5.2     tidyselect_1.2.1
## [5] splines_4.5.2     scales_1.4.0      yaml_2.3.12        fastmap_1.2.0
## [9] lattice_0.22-7    R6_2.6.1          labeling_0.4.3     generics_0.1.4
## [13] knitr_1.51        backports_1.5.0   pillar_1.11.1      RColorBrewer_1.1-3
## [17] tzdb_0.5.0        rlang_1.1.7       stringi_1.8.7      xfun_0.56
## [21] S7_0.2.1          otel_0.2.0        timechange_0.4.0   cli_3.6.5
## [25] mgcv_1.9-3        withr_3.0.2       magrittr_2.0.4     digest_0.6.39
## [29] grid_4.5.2        rstudioapi_0.18.0 hms_1.1.4          nlme_3.1-168
## [33] lifecycle_1.0.5   vctrs_0.7.0       evaluate_1.0.5     glue_1.8.0
## [37] farver_2.1.2      rmarkdown_2.30    tools_4.5.2        pkgconfig_2.0.3
## [41] htmltools_0.5.9
```

10. References

- Dataset documentation.

College.rds dataset

<https://docs.google.com/spreadsheets/d/1YrYiJ-J9rLhs4Qf5OCvHGMyUFyMKOMtMuGmpyt47mEY/edit?gid=1593529489#gid=1593529489>